Future Scenarios for AI-Powered Clinical Decision Support Systems

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Within the next 5–10 years, AI-powered Clinical Decision Support Systems (CDSS) are expected to become a standard component of clinical practice, significantly enhancing diagnostic accuracy, treatment planning, and workflow efficiency. In this future scenario, advanced CDSS platforms will integrate large language models (LLMs), machine learning, and predictive analytics directly into electronic health record (EHR) systems. These tools will analyze both structured data, such as laboratory values and vital signs, and unstructured data, including clinical notes and imaging reports, to generate real-time differential diagnoses and evidence-based care recommendations.

Healthcare professionals will interact with AI-powered CDSS through clinician-facing interfaces that provide contextual alerts, summarized clinical insights, and draft care plans. Systems similar to Glass Health will support, rather than replace, clinical judgment by allowing providers to review, validate, and modify AI-generated outputs before applying them to patient care. This collaborative interaction will promote trust while preserving clinician autonomy (Sendak et al., 2024).

The widespread adoption of AI-powered CDSS has the potential to improve patient outcomes by enabling earlier detection of disease, reducing diagnostic errors, and supporting personalized treatment strategies. Additionally, by automating routine documentation and cognitive tasks, these systems may reduce clinician burnout and improve care efficiency (Rajpurkar et al., 2023). However, ethical challenges remain, including algorithmic bias, data privacy, transparency, and regulatory compliance. Addressing these concerns will require explainable AI models, diverse training datasets, and adherence to established regulatory

frameworks such as FDA guidance for clinical decision support software (U.S. Food and Drug Administration [FDA], 2023).

Compared to current rule-based CDSS, which primarily rely on static alerts and predefined guidelines, future AI-powered systems will be adaptive and continuously learning from new clinical data and research. Overall, responsibly implemented AI-powered CDSS have the potential to significantly enhance healthcare quality, safety, and sustainability.

References

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